

Topic Overview: Connecting AWS S3 to Snowflake MANUAL AND AUTOMATION PROCESS

Snowflake is a cloud data warehouse, and AWS S3 is a cloud storage service. Often, data is stored in S3 and needs to be loaded into Snowflake for analytics. There are two main approaches to do this:

1. **Manual approach** – step-by-step commands and processes you run yourself.
 2. **Automation approach** – automated pipelines that run without manual intervention.
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AWS S3 → SNOWFLAKE

MANUAL PROCESS (UI-BASED)

STEP 1: Create S3 Bucket (AWS – Manual)

How (UI)

1. Login to **AWS Console**
2. Go to **S3 → Create Bucket**
3. Enter bucket name
Example: my-practice-bucket
4. Create folder:
5. employee_data/
6. Upload file:
7. emp_dec_2025.csv

✓ File is now available in cloud

STEP 2: Create FILE FORMAT (Snowflake UI)

How (UI)

1. Login to **Snowflake Web UI**
2. Go to **Databases**
3. Select your **Database → Schema**
4. Click **File Formats**
5. Click **Create**

Fill Details

Field	Value
Name	ff_emp_csv
Type	CSV
Skip Header	1
Field Enclosed By	"
Null If	NULL
Empty Field As Null	True

✓ Click **Create**

✓ STEP 3: Create EXTERNAL STAGE (Snowflake UI)

How (UI)

1. Go to **Database → Schema**
2. Click **Stages**
3. Click **Create → Stage**
4. Select **External Stage**

Fill Details

Field	Value
Stage Name	stg_emp_s3
URL	s3://my-practice-bucket/employee_data/
File Format	ff_emp_csv
Storage Integration	(select if available)

✓ Click **Create**

✦ This step **only creates a connection**

✅ STEP 4: Validate STAGE (Manual)

How (UI)

1. Open the created **Stage**
2. Click **Preview** or **Files**

✓ If file is visible → stage is correct

✗ If empty → path/permission issue

(Optional SQL check)

```
LIST @stg_emp_s3;
```

✅ STEP 5: Create TARGET TABLE (Snowflake UI)

How (UI)

1. Go to **Tables**
2. Click **Create** → **Table**
3. Choose **Standard Table**
4. Add columns:

Column	Type
emp_id	INT
emp_name	STRING
dept	STRING
salary	NUMBER(10,2)
join_date	DATE

✓ Click **Create Table**

✅ STEP 6: COPY INTO (Manual Load)

How (UI)

1. Open the **target table**
2. Click **Load Data**
3. Choose **Stage**
4. Select:
 - Stage: stg_emp_s3
 - File: emp_dec_2025.csv
5. Set **ON_ERROR → CONTINUE**
6. Click **Load**

✓ Data loaded successfully

● ON_ERROR (Manual Meaning)

Option	Meaning
CONTINUE	Load good rows, skip bad rows
ABORT_STATEMENT (default)	Stop load if error occurs
SKIP_FILE	Skip entire file if any error
SKIP_FILE_10	Skip file if >10 errors

✦ Practice = CONTINUE

WHY CONTINUE IS USED IN PRACTICE

- Do not fail entire load
 - Helpful during learning & testing
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MANUAL FLOW MEMORY

AWS S3 (Upload File)



Snowflake File Format (UI)



External Stage (UI)



Preview Stage



Create Table



Load Data (ON_ERROR)

INTERVIEW SAFE LINE (MANUAL)

“Using Snowflake UI, we manually create file formats and external stages, validate the stage, create target tables, and load data from S3 using COPY with ON_ERROR handling.”

■ AWS S3 → SNOWFLAKE

AUTOMATION PROCESS (Writing SQL query)

✅ STEP 1: Create S3 Bucket (AWS)

What

An **S3 bucket** is cloud storage where we keep our data files (CSV).

Why

Snowflake **cannot load files from local system**.

So we upload files to **AWS S3 first**.

Example

my-practice-bucket/

└─ employee_data/

 └─ emp_dec_2025.csv

✦ This path will be used in **external stage**.

✅ STEP 2: Create FILE FORMAT

What

A **file format** tells Snowflake **how to read the file**.

Why

- Avoid repeating CSV settings
- Reusable
- Cleaner COPY command

Query

```
CREATE OR REPLACE FILE FORMAT ff_emp_csv
```

```
TYPE = 'CSV'
```

```
FIELD_OPTIONALLY_ENCLOSED_BY = ''
```

```
SKIP_HEADER = 1
```

```
NULL_IF = ('NULL','null')
```

```
EMPTY_FIELD_AS_NULL = TRUE;
```

Explain in Interview

“File format defines file type, header, delimiter, and null handling.”

✅ STEP 3: Create EXTERNAL STAGE

What

An **external stage** is a **pointer to S3 location**.

Why

Snowflake **loads data only through stages**.

Query

```
CREATE OR REPLACE STAGE stg_emp_s3
```

```
URL = 's3://my-practice-bucket/employee_data/'
```

```
FILE_FORMAT = ff_emp_csv;
```

✖ No data loaded yet – only connection created.

✅ STEP 4: Validate STAGE

What

Check whether Snowflake can **see files in S3**.

Why

- Avoid COPY failure
- Confirm path is correct

Query

```
LIST @stg_emp_s3;
```

Result

emp_dec_2025.csv

✓ Stage working

✖ Empty → path/permission issue

✅ STEP 5: Create TARGET TABLE

What

Target table stores loaded data.

Why

COPY INTO needs a destination table.

Query

```
CREATE OR REPLACE TABLE employee_data (
```

```
  emp_id INT,
```

```
  emp_name STRING,
```

```
dept STRING,  
salary NUMBER(10,2),  
join_date DATE  
);
```

📌 Column order should match CSV (basic practice rule).

✅ STEP 6: COPY INTO with ON_ERROR

What

This step **actually loads data**.

Why

Moves data from **S3** → **Snowflake table**.

Query

```
COPY INTO employee_data  
FROM @stg_emp_s3  
ON_ERROR = 'CONTINUE';
```

Option

Meaning

CONTINUE	Load good rows, skip bad rows
ABORT_STATEMENT (default)	Stop load if error occurs
SKIP_FILE	Skip entire file if any error
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📌 **Practice = CONTINUE**

WHY CONTINUE IS USED IN PRACTICE

- Do not fail entire load
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🧠 ONE-SHOT MEMORY FLOW

S3 Bucket



File Format



External Stage



LIST (Validate)



Target Table



COPY INTO (ON_ERROR)

INTERVIEW READY ANSWER (FINAL)

“We load AWS S3 data into Snowflake by creating a file format, external stage, validating the stage, creating the target table, and using COPY INTO with ON_ERROR handling.”